

Sec

29/10/28

Factor model for population

$$\text{Factor model} \begin{cases} \Sigma = LL^T + \Psi \\ \downarrow \\ X \text{ not available.} \end{cases} \quad \underline{X} = \mu + L\underline{F} + \underline{e}$$

Sample factor fitting (factor model solⁿ through sampling)

S_{xx} available:

$$S = \hat{L}\hat{L}^T + \hat{\Psi}$$

$\hat{L}$ → estimate for L .

$\hat{\Psi}$ → estimate for Ψ

$$\hat{\Psi} = \text{diag}(S - \hat{L}\hat{L}^T)$$

↳ Remove off-diagonal elements.

Optimizing? $\rightarrow$ PCA
 $\rightarrow$ MLE

$y \sim N(\mu, \Sigma)$
 random sample $\rightarrow x_1, x_2, \dots, x_n$

$$L(\mu, \Sigma, x_1, x_2, \dots, x_n) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{n/2}} \exp\left(-\frac{n}{2} (\text{trace}(\Sigma^{-1} S_n) + (\bar{x} - \mu)^T \Sigma^{-1} (\bar{x} - \mu))\right)$$

where $S_n = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^T$

Recall from notes: MLE of $\mu = \hat{\mu} = \bar{x}$
 MLE of $\Sigma = \hat{\Sigma} = S_n$

$$L(\hat{\mu}, \Sigma, x_1, x_2, \dots, x_n) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{n/2}} \exp\left(-\frac{n}{2} (\text{trace}(\Sigma^{-1} S_n))\right)$$

In the factor model, $\Sigma = LL^T + \Psi$.

Taking log $\Rightarrow \ell(\Sigma) = \log L(\hat{\mu}, \Sigma, x_1, \dots, x_n)$
 $= -\frac{n}{2} (\log |\Sigma| + \text{trace}(\Sigma^{-1} S_n)) + \text{constant}$
 $\Sigma = LL^T + \Psi$

constant can be ignored in maximizing process.

$\Rightarrow \max_{L, \Psi} \ell(\Sigma)$
 s.t. $\Sigma = LL^T + \Psi$

$$\Rightarrow \max_{L, \Psi} \left[-\frac{n}{2} (\log |LL^T + \Psi| + \text{trace}((LL^T + \Psi)^{-1} S_n)) \right]$$

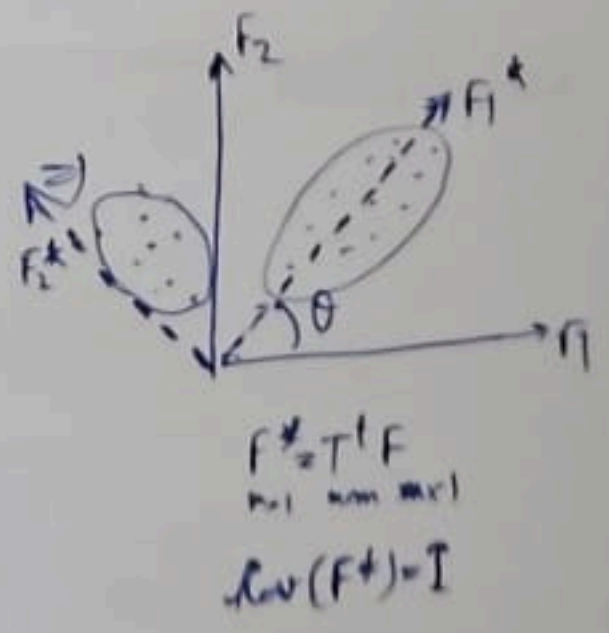
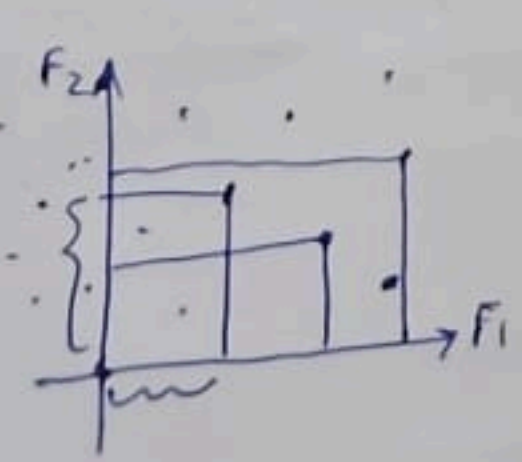
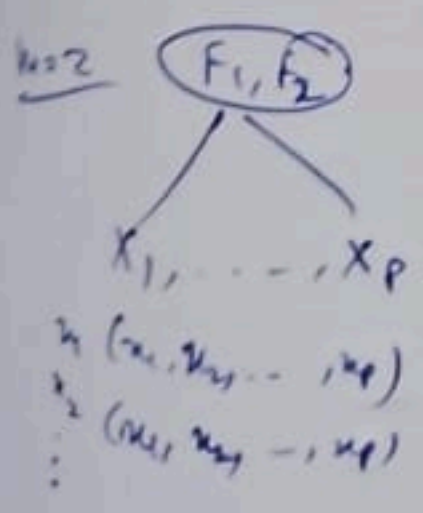
$\hookrightarrow$ not easy to solve by standard calculus.
 $\Rightarrow$ One has to adopt some numerical optimization techniques to find L & Ψ .
 $\hookrightarrow$ uniqueness is an issue here, because $L^* = LT$

T : orthonormal matrix is also a solution.

$$\hookrightarrow \max_{L, \Psi} \left[-\frac{n}{2} (\log(\dots) + \text{trace}(\dots)) \right]$$

s.t. $L^T \Psi^{-1} L = \Delta_{\text{diag}}$ for rotational invariance (fixes rotation)

Δ is a diagonal matrix (non-zero)



$$L^* T \Psi^{-1} L^* = (L^* T)^T \Psi^{-1} (L^* T)$$

$$= T^T L^T \Psi^{-1} L T \rightarrow \text{need not be diagonal}$$

$$= T^T \Delta T \cong \Delta$$

→ 2 matrices A & B are similar if $A = P^T B P$

↳ elementary row & column operations

Eigenvalues same, eigenvectors rotated by some T .